AQUAFAB INDUSTRIES

# Analysing and Exporting Results

Comparing treatments, recomputing from chosen cycles, and exporting what you publish.

For Aquafab Workbench 0.1.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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# 1 Analysing and exporting results

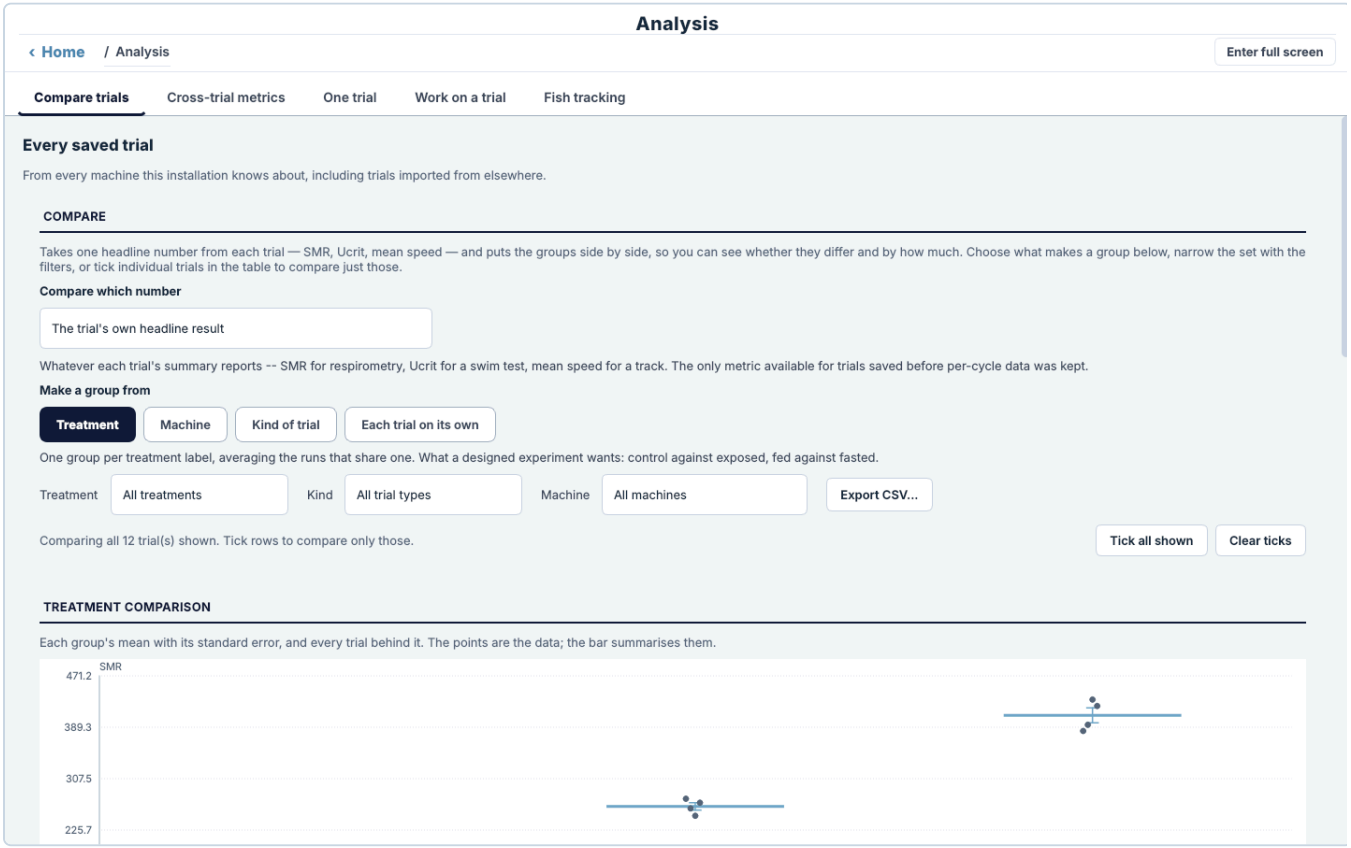
Analysis turns saved trials into the numbers you publish. It needs a license. It is in two places, which are the same view:

- the **Analysis** card on the Launcher: every trial from every machine on this computer, including imported ones;
- a machine's **Analysis** section: that machine's trials only.

> **Analysis never changes a trial.** Each trial keeps the result it reported when saved. Figures recomputed here are shown beside it, and say so.

> The screenshots show modelled example data made for this guide, not measurements.

## 1.1 Compare treatments




*Compare trials.*

**Compare trials** takes one number from each trial and puts groups side by side.

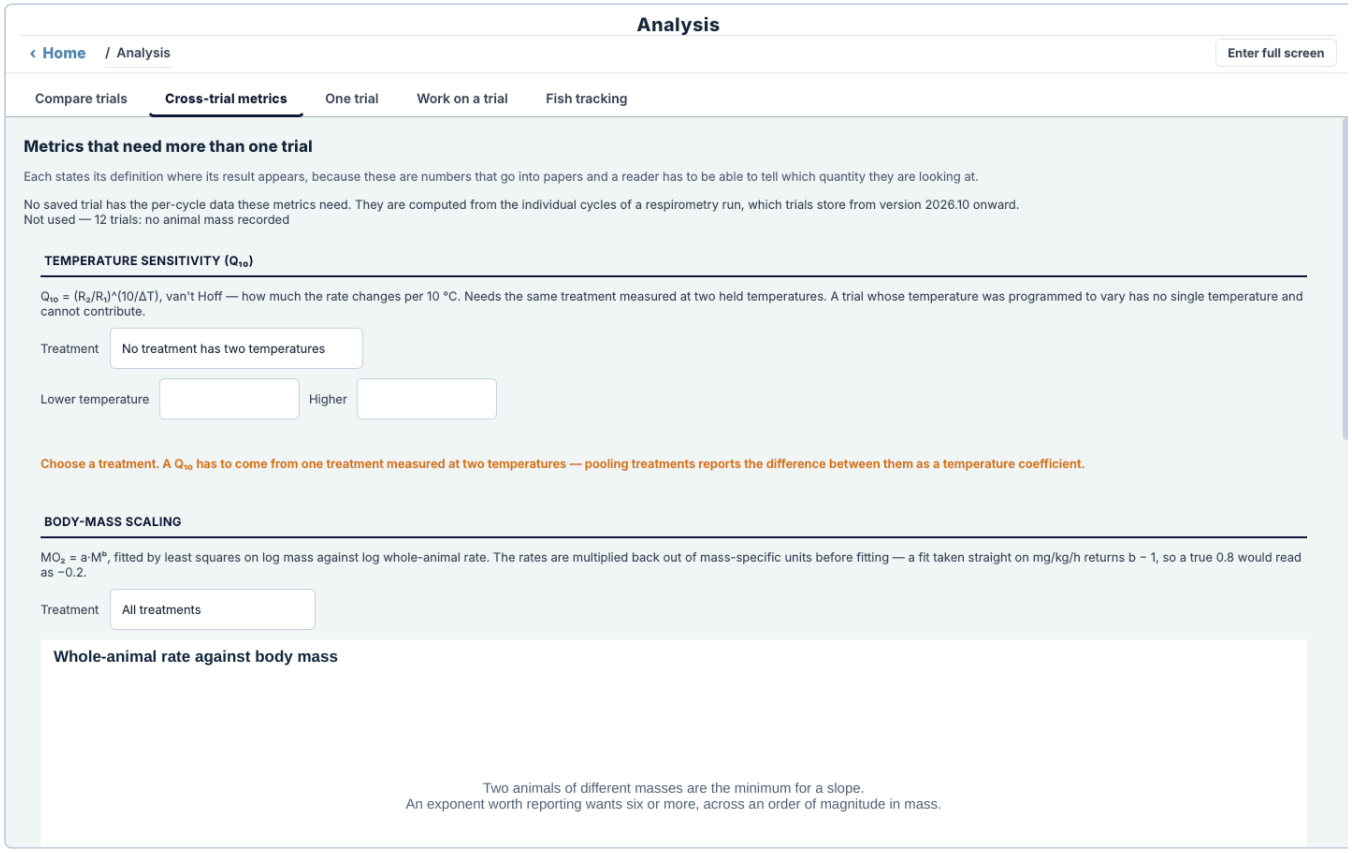
1. **Compare which number**: *The trial's own headline result* (SMR, $U_{crit}$ or mean speed), or, for respirometry trials, a specific metric: SMR by the trial's method or re-estimated, MMR, aerobic scope (absolute or factorial), mean or median $MO_2$, CV, whole-animal SMR, mass, oxygen levels, mean $R^2$, accepted and rejected cycles. The definition is shown under the menu.
2. **Make a group from**: **Treatment** (the usual choice), **Machine**, **Kind of trial**, or **Each trial on its own**.
3. Narrow with the treatment, **Kind** and **Machine** filters, or tick trials in the **Trials** table to compare only those.

What you get:

- **Treatment comparison**: each group's mean ± SE, with every trial as a point.
- **Traces, overlaid**: one line per trial ($MO_2$ or oxygen against cycle, $MO_2$ against oxygen, or fit quality). Never averaged together.
- **Group statistics**: n, mean, SD, SE and range. SD is how variable the animals were; SE is how well the mean is known. Compare treatments by SE.
- **Every parameter, every trial**: one row per trial, one column per parameter. A blank means it could not be computed, not zero.

**Export CSV...** writes the visible trials, one row each.

## 1.2 Cross-trial metrics



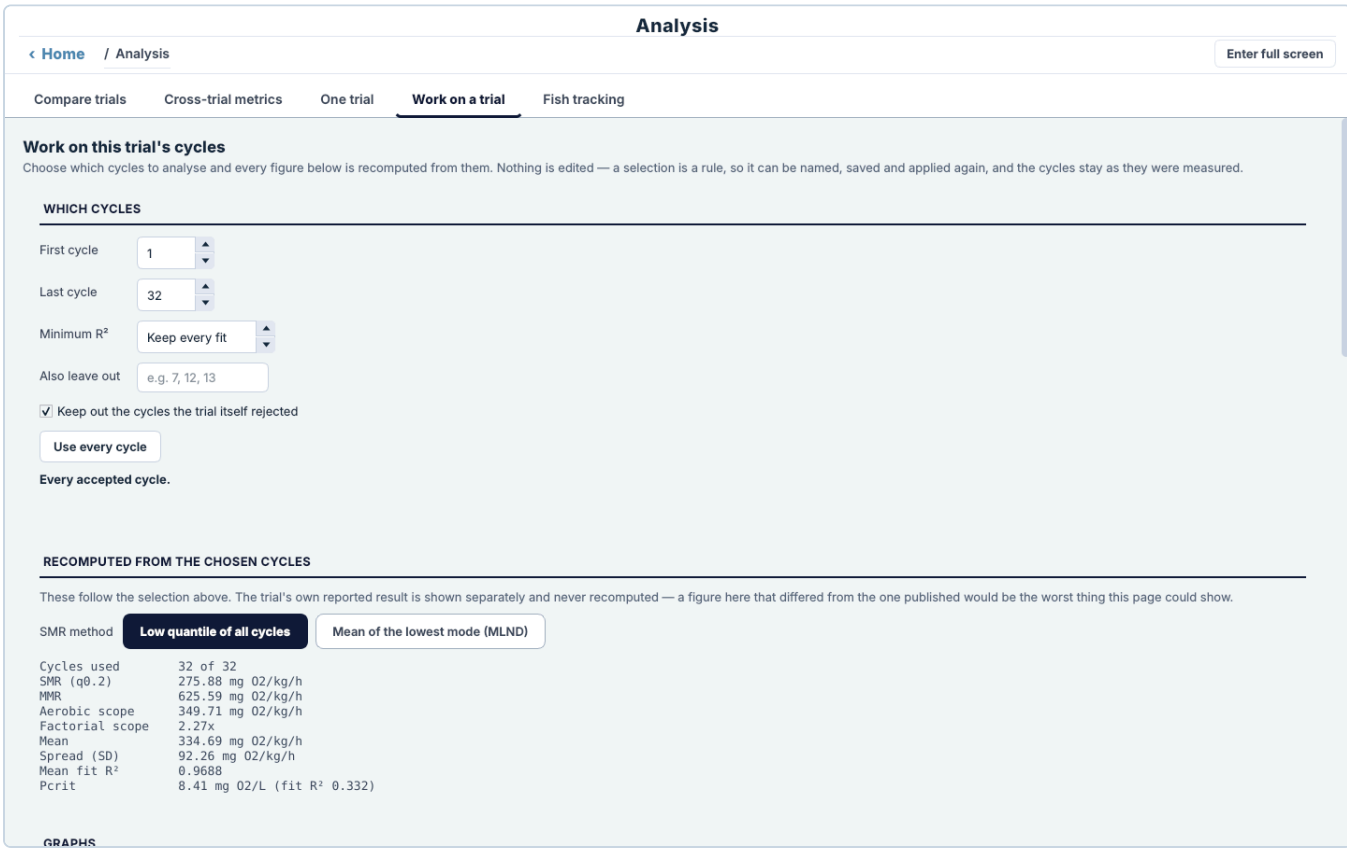

*Cross-trial metrics.*

| Metric | Needs |
|---|---|
| **$Q_{10}$**, $(R_2/R_1)^{(10/\Delta T)}$ | One treatment measured at two held temperatures. Choose the treatment and the two temperatures. |
| **Body-mass scaling**, $MO_2 = a \cdot M^b$ | At least two animals of different masses. Fitted on whole-animal rates, so b is the true exponent. |
| **What the SMR estimate came from** | A histogram of every accepted cycle, with the SMR marked. A disturbed animal shows as a high tail. |
| **Oxygen at which aerobic scope has halved** | A closed-chamber run where oxygen falls through the trial. |

## 1.3 Look at one trial

**One trial** (or **Inspect** in Trial History) shows everything about a run: the reported result exactly as saved, every measurement cycle with its $MO_2$, slope, $R^2$ and whether it was used, every setting the trial ran with, and notes. **Export CSV...** writes its cycles; **Reopen settings** loads its setup to run again.

## 1.4 Re-analyse with chosen cycles




*Work on a trial.*

**Work on a trial** recomputes everything from the cycles you choose. The trial itself is not edited.

1. Choose the trial.
2. **First cycle**: raise it past the cycles where the animal was settling. **Last cycle**, **Minimum $R^2$**, and **Also leave out** (for example *7, 12*) remove others. **Keep out the cycles the trial itself rejected** is on by default.
3. Choose the **SMR method**: *Low quantile* or *MLND*.
4. Read the recomputed SMR, MMR, scope, spread and, where oxygen declined, $P_{crit}$. In the graphs, hollow points are the cycles left out; hover for why.
5. Name the selection (*Settled cycles only*) and press **Save this selection**, so the figure can be reproduced. **Apply selected** brings one back.

## 1.5 Everything you can export

| What | Where |
|---|---|
| A trial's cycles (CSV) | **Export CSV…** on the trial page, or on the Respirometry Live view with the full oxygen trace |
| A report (PDF) | **Export PDF…** on Respirometry Live, or **Export PDF…** on Swim Performance |
| A comparison (CSV) | **Export CSV…** on Compare trials |
| A chart's data (CSV) | **Export…** on any chart's toolbar: the visible window and series |
| A track (CSV) | **Export Track…** on Fish Tracking |
| A vector field (CSV) | **Export Vector Field…** on Flow Field (PIV) |

## 1.6 Methods to cite

SMR: Chabot, Steffensen & Farrell (2016), *J. Fish Biol.* 88, 81–121. Background correction and intermittent flow: Svendsen, Bushnell & Steffensen (2016), *J. Fish Biol.* 88, 26–50. $U_{crit}$: Brett (1964), *J. Fish. Res. Bd. Canada* 21, 1183–1226. Oxygen solubility: García & Gordon (1992), *Limnol. Oceanogr.* 37, 1307–1312.